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Key Points:

- The study offers new dimensions for understanding and directing the psychological-social predictors of water-use behaviors of households
- Household water-use behavior is associated with three necessary conditions: capability, opportunity, and motivation (COM)
- COM dimensions are useful for the identification of behaviors that influence water-use and how these may diverge depending on the water-use character of the region and environment

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Household Water Use and Conservation Behavior: A Meta-Analysis

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Abstract Understanding the psychological-social drivers of water-use behavior in households is essential for enhancing the effectiveness of water-conservation strategies and subsequent environmental benefits. This study used the Behaviour Change Wheel framework to review associations between capability, opportunity, and motivation (COM) dimensions and household water-use behaviors. A meta-analysis of 88 correlation coefficients from a combined sample of 15,656 participants showed positive relationships between water-use behavior and COM dimensions. These three dimensions were statistically significant in predicting household water-use behavior, with opportunity being the most moderate predictor of water-conservation behavior ($r = 0.25, p < 0.001$), followed by motivation ($r = 0.24, p < 0.001$) and then capability ($r = 0.18, p < 0.001$). Collectively, these dimensions explained 37% of the variance in household water-conservation behavior. Correlation coefficients also diverged as a function of COM dimension subtypes (psychological, physical, social, reflective, and automatic) and study location, study design, and the gender of participants. Overall, the results are consistent with the Behaviour Change Wheel assertion that the integrative components of behavior are important sources of psychological-social drivers of water-use behavior. COM dimensions are useful for the identification of behaviors that influence water-use and how these may diverge depending on the water-use character of the region and environment.

1. Introduction

Water conservation by households commonly involves behaviors to reduce water consumption. These behaviors are generally enacted through curtailment or efficiency activities (Abrahamse & Steg, 2009; Fielding et al., 2012). Curtailment includes fixing leaking taps, reusing grey water, taking showers rather than baths, turning off taps while brushing teeth, and using dishwashers and washing machines with full loads (De Loë et al., 2001; Jorgensen et al., 2013). Efficiency requires greater personal involvement, financial investment, and technical know-how. It involves the use of water-efficient services such as low-volume shower heads, front-loading washing machines, water-efficient dishwashers, dual-flush toilets, and automatic reticulation systems (Richetin et al., 2016; Willis et al., 2011). Household water-use activities are influenced by different opportunities, capabilities, and motivations (Fielding et al., 2012; Michie et al., 2011), and knowledge of the attitudes toward, and the uptake of, these activities is important for effective water conservation policy and planning.

Models of human behavior provide a basis to examine variable-effect relationships in natural resource conservation behavior. The Theory of Planned Behavior (Ajzen, 1985) was used to propose that an individual's conservation behavior can be explained by behavioral intentions and environmental factors, which can either limit or facilitate actual control over a behavior (Ajzen, 1991). Intentions are conceived as a product of an individual's attitudes toward the behavior, social norms, and the perception of the degree of control they have to enact the behavior (Ajzen, 1991). The Theory of Reasoned Action (Ajzen & Fishbein, 1999) has shown that antecedent factors contribute to environmental attitudes (Fishbein & Ajzen, 2010). However, these applications of models of human behavior in natural resource conservation generally examine simple variable-effect relationships (Fishbein & Ajzen, 2010; Kollmuss & Agyeman, 2002; Steg et al., 2012; Triandis, 1977). Knowledge of individual variable-effect relationships have reduced household resource consumption by up to 21% in some studies (Randolph & Troy, 2008; Winett et al., 1982) but were not effective in sustaining long-term behavior change (World Health Organisation, 1996). Understanding causal mechanisms of behavior is required to achieve long-term water-conservation behavior change.

The adoption of household water-conservation behaviors is influenced by many psychological-social factors (Fielding et al., 2012). Communication is used to deliver water-conservation intervention strategies (Byrnes

et al., 2010). Communication strategies attempt to persuade households to adopt constant, operative, or intended actions to conserve water resources (Seyranian et al., 2015). Thus, communication can change behavior by challenging the view that wasting water is appropriate (Geller et al., 1983; Jorgensen et al., 2009). Cultural beliefs about water use can also influence the adoption of water-conservation behaviors (Jorgensen et al., 2009). Cultural beliefs affect water use through awareness of the value of water, and the benefits of conserving water. In similar environmental settings, diverse social groups may respond differently to the detrimental repercussions of water crises, in line with their prevailing belief systems and the values and norms of the group (Adger et al., 2012; Jones & Boyd, 2011; Moser & Ekstrom, 2010). Cultural traditions may also impede the implementation of adaptations that decrease susceptibility to drought and other external water threats (Nielsen & Reenberg, 2010). Understanding cultural beliefs about water wastage has been valuable for water utilities and individuals in reducing water use (Gilli, 2004).

Age and education are also psychological-social drivers of water-conservation behavior (Arbuthnott, 2009; Stern, 2000). Clark and Finley (2007) claimed older people are more likely to conserve water, although Kantola et al. (1982) found that older people are less likely to have sustained conservation intentions over longer periods of time. While retirees are greater water consumers (Lyman, 1992), teenage water consumption has been increasing since the 1980s (Mayer et al., 1999). Individuals with greater educational attainment have greater concern for the environment and are able to carry out informed and sustainable environmental practices (Arbuthnott, 2009; Fien et al., 2001; Tilbury et al., 2002). Other psychological-social drivers of the adoption of water-conservation behavior include household income (Gregory & Di Leo, 2003), type of home (Gilg & Barr, 2006), environmental beliefs or ecological worldview (Clark & Finley, 2007), and water pricing (Van Vugt, 2001).

A meta-analysis of the causal mechanisms of water-conservation behavior is presented in this manuscript. The Behaviour Change Wheel (BCW) framework of Michie et al. (2011) is used to organize psychological-social drivers and their effects on water conservation. These drivers were categorized into a broad integrative model of behavior comprising capability, opportunity, and motivation (COM) dimensions. Use of the COM dimensions to examine the causal mechanisms of water-conservation behavior link drivers of, and barriers to, behavioral change with intervention strategies and policies, which may result in long-term water conservation behavior.

2. Theoretical Framework

The BCW (Michie et al., 2011) provides a conceptual framework to examine effective behavior-change interventions for household water conservation. The BCW is an integrative framework that recognizes three conditions for behavior change: COM-behavior (the COM-B system). Behavior is conceptualized as a series of interactions that influence each individual condition; thus, behavior becomes part of an interacting system involving all conditions (Michie et al., 2011). For example, motivation is influenced by opportunity and capability, whereas enacting behavior can alter capability, motivation, and opportunity. Interventions designed to elicit behavioral change need to influence one or more of the individual conditions in order to enact the behavior. The COM-B system provides a mechanism for merging different but related behavior influences into a practical context that enables policy-makers to identify behaviors that are most relevant to household water use and to design behavior change interventions for water conservation.

The BCW allows for the identification of different causal mechanisms of behavior and for these to be placed into broad categories of COM. It does this by identifying applicable types of intervention strategies for behavior change, and the content and focus that an intervention requires. In particular, it identifies specific psychological-social mechanisms or environmental conditions linked to a behavior of interest. Bundling variables such as household size and water rate rebates into a single category (e.g., physical opportunity), serves to conflate the effects of quite different variables. The COM-B system has advantages over other approaches in designing intervention strategies. Intervention mapping (Bartholomew et al., 2011), for example, seeks to relate behavior to its theoretical cause(s). However, the BCW and COM-B system provide causal mechanisms of behavior that can link to interventions for behavior change (Michie et al., 2011).

The capability dimension of the COM-B system refers to the individual's physical or psychological ability to enact the behavior. It is characterized by two sub types. Physical capability is the degree to which an individual can physically enact or participate in a behavior and includes having the necessary physical

skills, strength, or stamina to change behavior (e.g., having the skills to repair minor water leakages). Psychological capability is an individual's knowledge and psychological skills to engage in the necessary mental processes to make an informed decision about a course of action (e.g., understanding the impact of water scarcity on the environment and reducing water wastage).

The opportunity dimension of the COM-B system is the external physical and social environment which enables the individual's behavior. Opportunity factors are generally external to an individual and make the behavior possible or prompt it. Opportunity is characterized by two sub types. Physical opportunity afforded by the environment can include time, resources and location, but also economic enablers of water conservation (e.g., being able to conserve water because of financial incentives or rebates to promote water-conservation measures). Social opportunity includes social cues and cultural norms that influence the way individuals think about making possible a behavior (e.g., being able to conserve water in the house because of feeling under social pressure to install water-efficient appliances in the house and garden). Trust is also part of social opportunity. People conserve water if they trust that others are minimizing their water use (i.e., inter-personal trust), and trust that water utilities and government are providing the necessary support for water conservation (i.e., institutional trust; Graymore & Wallis, 2010; Jorgensen et al., 2009).

The motivation dimension comprises factors intrinsic and extrinsic to an individual that enact or direct behavior. This dimension is characterized by two sub types. Reflective motivation involves intentions and evaluations (beliefs about what is good and bad, conscious intentions, decisions, and plans) of the behavior (e.g., a householder intending to take shorter showers). Automatic motivation involves emotional reactions, desires (wants/needs), inhibitions and reflex responses which activate or inhibit behavior (e.g., feeling good by thinking about conserving water for the next generation). Motivation broadly involves evaluation mechanisms of water-conservation behavior, general environmental attitudes, norms (personal and social), values, broad beliefs about the natural environment, water-peculiar beliefs and conventional practices pertaining to water use.

In summary, the BCW is a framework for linking behavioral conditions to specific types of behavior-change interventions and policies (Michie et al., 2011). Examples include environmental/social planning, fiscal measures, guidelines, communication/marketing, legislation, service provision, and regulation. The use of the BCW model and the COM-B system to examine the psychological-social drivers of water-conservation behavior can enhance the engagement of respondents in innovative ways based on the householders' COM for conserving water.

3. Materials and Methods

3.1. Literature Search and Screening

Two search approaches were used to find relevant studies for the meta-analysis. First, we systematically searched the PsycINFO, ProQuest, and JSTOR databases for studies written in English reporting behavioral drivers of household water-use. The following key terms related to household water use, the COM-B system, and water conservation were used: *water-efficient plumbing fixtures, water-conservation measures, purchase decision, cultural beliefs, environmental beliefs, social norms, perceived behavioral control, educational background, landscape choices, home ownership, type of house, dwelling context, moral persuasion, normative beliefs, awareness, conservation campaign, subjective norms, water policy, water pricing, water restrictions, commitment, turning off tap, perceived efficacy, water-use habits, financial incentives, proenvironmental behavior, communication*, and combinations of these truncated key terms. Second, water-conservation researchers were contacted directly to identify and obtain unpublished data sets and reports on behavior change related to water-conservation activities.

Standard methods were used to record, screen, and assess the eligibility of all identified articles and other information (Figure 1). Screening was done using the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) protocol, which is an evidence-based minimum set of items for reporting in systematic reviews and meta-analyses (Moher et al., 2009). The *metagear* package in R was used to filter abstracts and titles and to sift, flag, and remove duplicate studies (Lajeunesse, 2016). To be included in the review, studies must have assessed the relationship between at least one COM dimension and at least one water-conservation behavior outcome. Psychological-social predictors must have reported effect sizes or included enough information (e.g., mean, standard deviation, variance, *t* test, *F* test, percentage, or chi-

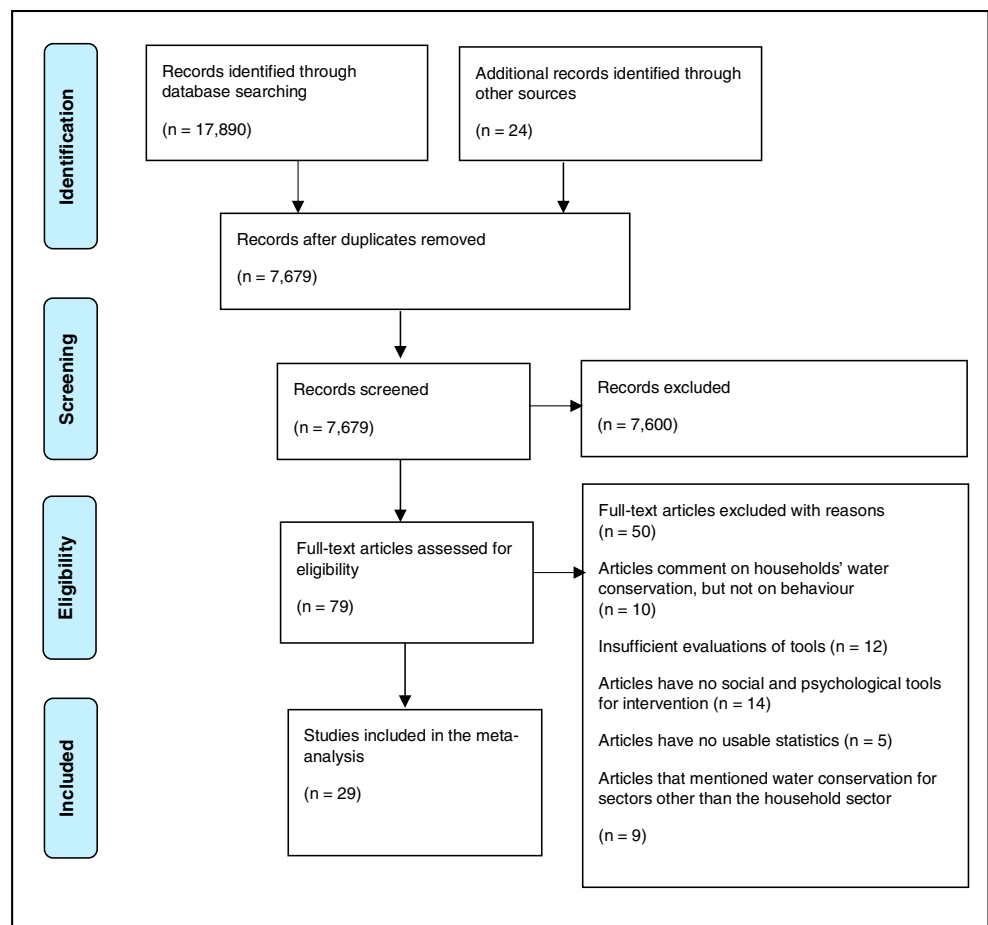


Figure 1. Flow of information used in the meta-analysis of household water-use. The flow included the identification, screening, eligibility, and inclusion phases of the literature review, following the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) method (Moher et al., 2009).

square) for effect sizes to be computed. The reference list of each retrieved article was further examined for relevant literature, and the Google Scholar search engine was also used to search for articles by identified authorities in the discipline. Many community-based social marketing and behavior-change studies driven by government initiatives contain relevant information, and these were included in the analysis. In total, 88 reported effect sizes ($k = 88$) were found that met the inclusion criteria (Table 1).

3.2. COM Dimensions and Data Aggregation

To create variables representing the COM dimensions, we identified latent psychological-social drivers in the literature. These were recorded and grouped to generate variables that represent the COM dimensions as per the COM-B system. We collapsed the data using an aggregate function in *R* to group variables (psychological-social predictors) into relevant COM dimensions. The aggregate (scalar) function calculated summary statistics that made up the aggregated values. These aggregated values were further categorized under each COM dimension subtype to generate coding categories. Thus, each psychological-social predictor with computed effect size was segregated into the six subtypes of the COM dimensions for the meta-analysis.

3.3. Coding and Categorization

We adopted standard meta-analysis procedures to code and analyze the data from each article. Data were coded according to the information outlined in Table 1: (1) authors and year of publication, (2) water-use behavior outcome variables and COM dimensions and subtypes, (3) predictor variables that underscored the COM dimension subtypes, (4) methodological moderators, and (5) effect size (r) based on the Pearson's r statistic. The meta-analysis then computed average effect sizes by the diverse types of coded

Table 1
Summary of Articles Included in the Meta-analytic Review and the Variables Coded in Each Article

Study	N	Country	Sample type	Study design	Predictor variable	COM subtype	COM dimensions	Water-use behavior	% Females	r	SE
Adams et al. (2013)	2226	United States	General population	Longitudinal	Information	Psychological	Capability	Household water conservation	Unknown	0.42	0.02
Adams et al. (2013)	2226	United States	General population	Longitudinal	Attitude toward water conservation	Reflective	Motivation	Household water conservation	Unknown	0.07	0.02
Adams et al. (2013)	2226	United States	General population	Longitudinal	Location	Physical	Opportunity	Household water conservation	Unknown	0.02	0.02
Adams et al. (2013)	2226	United States	General population	Longitudinal	Time	Physical	Opportunity	Household water conservation	Unknown	0.16	0.02
Adams et al. (2013)	2226	United States	General population	Longitudinal	Education	Psychological	Capability	Household water conservation	Unknown	0.01	0.02
Adams et al. (2013)	2226	United States	General population	Longitudinal	Knowledge	Psychological	Capability	Household water conservation	Unknown	0.34	0.02
Aitken et al. (1994)	321	Australia	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water conservation	Unknown	0.30	0.05
Aitken et al. (1994)	321	Australia	General population	Correlational	Habits	Automatic	Motivation	Household water efficiency	Unknown	0.24	0.05
Aitken et al. (1994)	321	Australia	General population	Correlational	Values	Reflective	Motivation	Household water efficiency	Unknown	0.31	0.05
Bernedo et al. (2014)	400	United States	General population	Experimental	Social norms	Social	Opportunity	Household water efficiency	Unknown	0.02	0.05
Chang et al. (2013)	900	China	General population	Correlational	Environmental concern	Psychological	Capability	Household water conservation	44.60	0.11	0.03
Chang et al. (2013)	900	China	General population	Correlational	Normative beliefs	Social	Opportunity	Household water conservation	44.60	0.05	0.03
Chang et al. (2013)	900	China	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water conservation	44.60	0.06	0.03
Chang et al. (2013)	900	China	General population	Correlational	Social norms	Social	Opportunity	Household water conservation	44.60	0.12	0.03
Chang et al. (2013)	900	China	General population	Correlational	Information	Psychological	Capability	Household water conservation	44.60	0.06	0.03
Clark & Finley (2007)	750	Bulgaria	General population	Longitudinal	Attitude toward water conservation	Reflective	Motivation	Household water conservation	Unknown	0.53	0.03
Clark & Finley (2007)	750	Bulgaria	General population	Longitudinal	Environmental concern	Psychological	Capability	Household water-saving activities	Unknown	0.21	0.04
Clark & Finley (2007)	750	Bulgaria	General population	Longitudinal	Information	Psychological	Capability	Household water-saving activities	Unknown	0.10	0.04
Corral-Verdugo and Frías-Armenta (2006)	177	Mexico	General population	Correlational	Social norms	Social	Opportunity	Household water-saving activities	44.60	−0.18	0.07

Table 1
(continued)

Study	N	Country	Sample type	Study design	Predictor variable	COM subtype	COM dimensions	Water-use behavior	% Females	r	SE
Corral-Verdugo et al. (2003)	510	Mexico	General population	Longitudinal	Normative beliefs	Social	Opportunity	Household water conservation	61.00	0.22	0.04
Dascher et al. (2014)	273	United States	General population	Longitudinal	Information	Psychological	Capability	Household water efficiency	64.10	0.41	0.05
Dascher et al. (2014)	273	United States	General population	Longitudinal	Water restrictions	Physical	Opportunity	Household water conservation	64.10	0.32	0.06
Dascher et al. (2014)	273	United States	General population	Longitudinal	Rebate	Physical	Opportunity	Household water conservation	64.10	0.01	0.06
De Oliver (1999)	203	United States	General population	Correlational	Income	Physical	Capability	Household water conservation	Unknown	−0.07	0.07
De Oliver (1999)	203	United States	General population	Correlational	Education	Psychological	Capability	Household water conservation	Unknown	0.05	0.07
Dickerson et al. (1992)	80	United States	Students	Experimental	Mindfulness	Reflective	Motivation	Household water conservation	100.00	0.59	0.07
Dickerson et al. (1992)	80	United States	Students	Experimental	Commitment	Reflective	Motivation	Household water-use activities	100.00	0.42	0.09
Dickerson et al. (1992)	80	United States	Students	Experimental	Household size	Physical	Opportunity	Household water-saving activities	100.00	0.09	0.11
Domene & Saurí (2006)	532	Spain	General population	Correlational	Income	Physical	Capability	Household water-saving activities	Unknown	0.01	0.04
Domene & Saurí (2006)	532	Spain	General population	Correlational	Household size	Physical	Opportunity	Household water efficiency	Unknown	0.09	0.04
Fan et al. (2014)	776	China	General population	Longitudinal	Perceptions of water-efficient devices	Psychological	Capability	Household water efficiency	Unknown	0.03	0.04
Fielding et al. (2012)	1008	Australia	General population	Correlational	Intentions	Reflective	Motivation	Household water efficiency	56.60	0.52	0.02
Fielding et al. (2012)	1008	Australia	General population	Correlational	Attitude toward conservation	Reflective	Motivation	Household water-saving activities	56.60	0.42	0.03
Fielding et al. (2012)	1008	Australia	General population	Correlational	Social norms	Social	Opportunity	Household water-saving activities	56.60	0.36	0.03
Fielding et al. (2012)	1008	Australia	General population	Correlational	Perceived behavioral control	Psychological	Capability	Household water-saving activities	56.60	0.27	0.03
Geller et al. (1983)	129	United States	General population	Correlational	Household size	Physical	Opportunity	Household water-saving activities	Unknown	0.40	0.08
Geller et al. (1983)	129	United States	General population	Correlational	Temperature	Physical	Opportunity	Household water efficiency	Unknown	0.30	0.08
Geller et al. (1983)	129	United States	General population	Correlational	Humidity	Physical	Opportunity	Household water efficiency	Unknown	0.25	0.08

Table 1
(continued)

Study	N	Country	Sample type	Study design	Predictor variable	COM subtype	COM dimensions	Water-use behavior	% Females	r	SE
Gilg & Barr (2006)	1600	UK	General population	Longitudinal	Education	Psychological	Capability	Household water efficiency	65.00	0.22	0.02
Gilg & Barr (2006)	1600	UK	General population	Longitudinal	Purchase decisions	Reflective	Motivation	Household water-saving activities	65.00	0.12	0.03
Gilg & Barr (2006)	1600	UK	General population	Longitudinal	Habits	Automatic	Motivation	Household water-saving activities	65.00	0.13	0.03
Gilg & Barr (2006)	1600	UK	General population	Longitudinal	Income	Physical	Capability	Household water-saving activities	65.00	0.26	0.03
Gilg & Barr (2006)	1600	UK	General population	Longitudinal	House ownership	Physical	Opportunity	Household water-saving activities	65.00	0.63	0.02
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Awareness	Psychological	Capability	Household water-saving activities	Unknown	−0.05	0.05
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Involvement	Physical	Capability	Household water efficiency	Unknown	−0.17	0.05
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water efficiency	Unknown	0.03	0.06
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Habits	Automatic	Motivation	Household water efficiency	Unknown	−0.24	0.04
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Age	Physical	Capability	Household water efficiency	Unknown	−0.34	0.04
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Income	Physical	Capability	Household water efficiency	Unknown	0.42	0.04
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Education	Psychological	Capability	Household water efficiency	Unknown	0.17	0.05
Gregory & Di Leo (2003)	471	Australia	General population	Correlational	Household size	Physical	Opportunity	Household water efficiency	Unknown	0.48	0.04
Hamilton (1983)	431	United States	General population	Correlational	Income	Physical	Capability	Household water efficiency	Unknown	0.26	0.05
Hamilton (1983)	431	United States	General population	Correlational	Education	Psychological	Capability	Household water conservation	Unknown	0.11	0.05
Hamilton (1983)	431	United States	General population	Correlational	Household size	Physical	Opportunity	Household water conservation	Unknown	0.16	0.04
Kurz et al. (2005)	166	Australia	General population	Experimental	Use of water-efficient devices	Physical	Capability	Household water conservation	Unknown	0.23	0.07
Kurz et al. (2005)	166	Australia	General population	Experimental	Attitude toward conservation	Reflective	Motivation	Household water efficiency	Unknown	0.18	0.08
Kurz et al. (2005)	166	Australia	General population	Experimental	Information	Psychological	Capability	Household water efficiency	Unknown	0.67	0.04

Table 1
(continued)

Study	N	Country	Sample type	Study design	Predictor variable	COM subtype	COM dimensions	Water-use behavior	% Females	r	SE
Kurz et al. (2005)	166	Australia	General population	Experimental	Feedback	Psychological	Capability	Household water efficiency	Unknown	0.47	0.06
Lam (2006)	166	Taiwan	General population	Experimental	Attitude toward conservation	Reflective	Motivation	Household water efficiency	50.00	0.24	0.07
Lam (2006)	166	Taiwan	General population	Experimental	Social norms	Social	Opportunity	Household water-saving activities	50.00	0.26	0.07
Lam (2006)	166	Taiwan	General population	Experimental	Perceived behavioral control	Psychological	Capability	Household water-saving activities	50.00	0.20	0.08
Lapinski et al. (2007)	72	United States	Students	Experimental	Social norms	Social	Opportunity	Household water-saving activities	49.00	0.52	0.09
Lapinski et al. (2007)	72	United States	Students	Experimental	Attitude toward water conservation	Reflective	Motivation	Household water conservation	49.00	0.25	0.11
Lapinski et al. (2007)	72	United States	Students	Experimental	Intentions	Reflective	Motivation	Household water conservation	49.00	0.28	0.11
Lowe et al. (2015)	909	Australia	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water conservation	59.00	0.38	0.03
Lowe et al. (2015)	909	Australia	General population	Correlational	Social norms	Social	Opportunity	Household water efficiency	59.00	0.24	0.03
Lowe et al. (2015)	909	Australia	General population	Correlational	Perceived behavioral control	Psychological	Capability	Household water efficiency	59.00	−0.03	0.03
Lowe et al. (2015)	909	Australia	General population	Correlational	Moral obligation	Reflective	Motivation	Household water efficiency	59.00	0.14	0.03
Lowe et al. (2015)	909	Australia	General population	Correlational	Trust	Social	Opportunity	Household water efficiency	59.00	0.23	0.03
Marandu et al. (2010)	462	Botswana	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water efficiency	54.00	0.23	0.04
Marandu et al. (2010)	462	Botswana	General population	Correlational	Social norms	Social	Opportunity	Household water conservation	54.00	0.24	0.04
Newton & Meyer (2012)	1250	Australia	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water conservation	Unknown	−0.01	0.03
Newton & Meyer (2012)	1250	Australia	General population	Correlational	Household type	Physical	Opportunity	Household water efficiency	Unknown	0.41	0.02
Newton & Meyer (2012)	1250	Australia	General population	Correlational	Landscape choices	Physical	Opportunity	Household water efficiency	Unknown	0.44	0.02
Newton & Meyer (2012)	1250	Australia	General population	Correlational	Location	Physical	Opportunity	Household water efficiency	Unknown	0.04	0.03
Richetin et al. (2016)	41	UK	Students	Experimental	Descriptive norms	Social	Opportunity	Household water efficiency	46.00	0.38	0.14

Table 1
(continued)

Study	N	Country	Sample type	Study design	Predictor variable	COM subtype	COM dimensions	Water-use behavior	% Females	r	SE
Seyranian et al. (2015)	374	United States	General population	Correlational	Social identity	Social	Opportunity	Household water-saving activities	Unknown	0.06	0.05
Seyranian et al. (2015)	374	United States	General population	Correlational	Personal identity	Social	Opportunity	Household water efficiency	Unknown	0.19	0.05
Seyranian et al. (2015)	374	United States	General population	Correlational	Social norms	Social	Opportunity	Household water efficiency	Unknown	0.15	0.05
Stewart et al. (2013)	151	Australia	General population	Longitudinal	Prompt	Reflective	Motivation	Household water efficiency	Unknown	0.27	0.08
Trumbo & O'Keefe (2005)	868	United States	General population	Correlational	Values	Reflective	Motivation	Household water efficiency	Unknown	0.16	0.03
Trumbo & O'Keefe (2005)	868	United States	General population	Correlational	Past actions	Physical	Capability	Household water conservation	Unknown	0.11	0.03
Trumbo & O'Keefe (2005)	868	United States	General population	Correlational	Social norms	Social	Opportunity	Household water conservation	Unknown	0.44	0.02
Trumbo & O'Keefe (2005)	868	United States	General population	Correlational	Attitude toward water conservation	Reflective	Motivation	Household water conservation	Unknown	0.40	0.03
Trumbo & O'Keefe (2005)	868	United States	General population	Correlational	Information	Psychological	Capability	Household water conservation	Unknown	0.40	0.03
Van Vugt (2001)	278	UK	General population	Longitudinal	Household size	Physical	Opportunity	Household water conservation	Unknown	0.33	0.05
Willis et al. (2011)	132	Australia	General population	Longitudinal	Environmental concern	Psychological	Capability	Household Water-saving activities	Unknown	0.14	0.09
Willis et al. (2011)	132	Australia	General population	Longitudinal	Perceptions of water-efficient devices	Psychological	Capability	Household water-saving activities	Unknown	0.22	0.08

Note. N = sample size, r = point estimate of the effect size (Pearson's r correlation coefficient) using the summary correlation of outcomes reported, SE = standard error.

variables (e.g., habits, water-efficient devices, norms, perceived behavioral control, and awareness). The three variables considered were as follows:

1. Outcome variables. Each study was coded for a specific outcome variable for water-use behavior, defined to include water efficiency, conservation, and water-saving activities. Water-use behavior involves actions, programs, and procedures carried out to conserve water resources to support the environment and to meet present and future use. It includes self-reports of past consumption behaviors (e.g., average number of showers per week), self-reports of efficiency and curtailment behaviors provided by a single household resident and using improved technology or devices to reduce water consumption (e.g., water-efficient washing machines and dishwashers). The studies were also coded for the COM dimensions (COM) and COM dimension subtypes (physical, psychological, social, reflective, and automatic).
2. Predictor variables. The studies in the meta-analysis measured various predictors in the context of COM influences on behavior, which included, for example, educational background, perceived behavioral control, water policy, cultural beliefs, subjective norms, and water-use habits (Table 1). Each predictor was coded within the COM dimension subtypes to assess associations between the COM dimension subtypes and the water-conservation outcome variables.

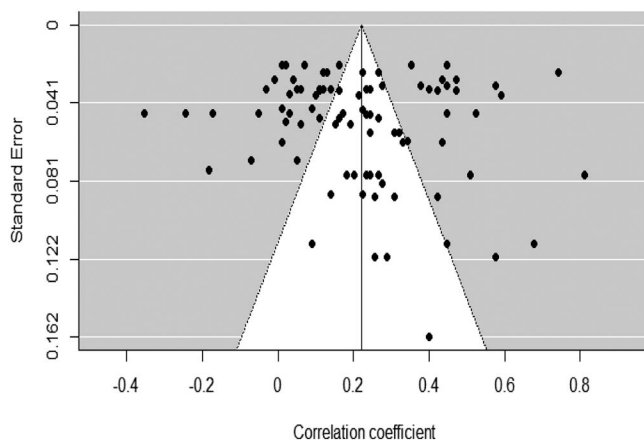


Figure 2. Funnel plot to illustrate potential publication bias and the effect sizes of the correlation between COM dimensions and household water-use behavior. Egger's regression test ($p = 0.122$), and the Rank correlation test ($p = 0.196$) were not statistically significant.

both intra-study and inter-study variability. In terms of study weighting, the random-effect model guided the study to assign larger weights to studies that involved a relatively smaller sample size, whereas smaller weights were given to larger studies. Employing the random-effects model provides larger confidence intervals resulting in higher conventional values and conclusions (Lipsey & Wilson, 2001). Computationally, the inverse variance of each study was used to individually weight each study (Benos & Zotou, 2014; Stanley & Doucouliagos, 2012). The meta-analysis was conducted in *R* using the Meta-Analysis package (*metafor*; Viechtbauer, 2010) and Robust Variance Meta-Regression package (*robumeta*). The *metafor* package performed the random-effect modeling, whereas the *robumeta* package was used to perform uncertainty analysis and produce funnel plots of effect sizes (Hedges et al., 2010). In the random-effect modeling, the restricted maximum-likelihood (REML) method was used given that it is negatively biased and reported to have the best properties in estimating variance components (Viechtbauer et al., 2015). The Pearson's r correlation coefficient was employed as the index of effect size in all analyses. This correlation coefficient is not normally distributed, so we converted the Pearson's r correlations to Fisher's z scores using Fisher's transformation and then converted Fisher's (z) back to Pearson's r for reporting the average correlation along with the 95% confidence interval, as recommended by Lipsey and Wilson (2001). Heterogeneity was assessed using the Q statistic (Hedges & Olkin, 1985), where a low p value is indicative of heterogeneity among the relevant effect sizes.

In those studies that presented multiple water-use behavior outcome measures, more than one effect size was coded. To meet the key statistical assumption of effect size independence (Lipsey & Wilson, 2001), we computed an average effect size for each water-use behavior outcome measure for each of these studies to obtain an overall correlation value. This ensured that an individual study presented a single effect size to each of the general analyses ($k = 88$) based on 15,656 participants.

3.5. Issues of Publication Bias

A point of contention in many meta-analyses is the likelihood for information bias, where the research outcomes depend not just on the nature of the research but on the hypothesis experimented with and the magnitude and direction of effects observed in the published literature (Dickersin, 1990; Rothstein et al., 2005). To examine potential publication bias in our study, we produced funnel plots for effect sizes by standard error and used the Egger's mixed-effects meta-regression model (Egger et al., 1997; Sterne & Egger, 2001) and a rank correlation test (Begg & Mazumdar, 1994) to quantify possible biases in the funnel plot using *metafor* and *robumeta*. The accuracy of each study (inversion of the standard error, y axis) was mapped out against its correlation coefficient (x axis). The funnel plot was a symmetrical inverted funnel, indicating no systematic publication bias (Figure 2). It also showed that the 88 effect sizes considered in this study were distributed symmetrically on both sides of the average effect sizes. To buttress for an improved evaluation accuracy, we further used Egger's regression intercept test in which the accuracy

3. Moderator variables. Methodological moderator variables were also coded to understand the modification of intensity or path of the correlational association among the predictor variables and the water-use behavior outcome variables (Rose et al., 2004). Gender, study design, study location, and COM dimension subtypes were used as the moderator variables to assess their influence on effect sizes across the studies.

Coding reliability was determined using the method of Cooper (2010). The primary coder coded all studies, and the secondary coder recoded an arbitrary collection of 45% of the initial article set. Inter-coder reliability for the included studies was 95% (Cohen's $k = 0.85$), indicating very high consistency. Disparities were determined through panel discussion between the coders with all agreed documents being further assessed to establish consistency with any agreed changes. Data were analyzed after reaching the final consensus.

3.4. Data Analysis

Variations in true effects existing between studies were captured via a random effect model (cf. Quintana, 2015). The random-effect model assessed

Table 2
Effect Sizes for COM Dimensions

COM dimension	<i>k</i>	<i>r</i>	95% CI		<i>z</i>	<i>p</i>
			Lower	Upper		
Capability	31	0.18	0.09	0.26	4.42	<0.001
Opportunity	32	0.25	0.18	0.33	6.92	<0.001
Motivation	25	0.24	0.17	0.31	5.97	<0.001

Note. *k* = number of effect sizes. *r* = point estimate of the effect size (Pearson's *r* correlation coefficient), using the summary correlation of outcomes reported, CI = confidence interval; the 95% lower and upper limits, *Z* = Fisher's transformation of the effect size, *p* < 0.001. COM, capability, opportunity, and motivation.

from 41 to 2,226, with a mean of 539.8 (*SD* = 93.6). Most studies were conducted in Australia (36.4%), followed by the United States (35.2%), UK (8.0%), and China (6.8%). In terms of study design, 56.8% of studies used a correlational design, 26.1% used a longitudinal design, and 17.1% used an experimental design. The studies also yielded a range of predictor types, which underscored COM within each COM dimension (Table 1).

4.1. Overall Effect Sizes for COM Dimensions

Significant, but weak, linear associations between COM dimensions and water-use behavior were recorded (Table 2). The overall weighted effect size (*r* = 0.25) for opportunity and household water-use behavior indicates that participants who scored higher on the opportunity dimension perceived that the physical or social environment enables water-conservation behavior. The effect size for motivation (*r* = 0.24) indicates reflective or automatic mechanisms activate and/or inhibit water-conservation behavior, while the effect size for capability (*r* = 0.18) indicates participants who scored higher on the capability dimension perceived a psychological or physical ability to enact water-use behavior.

Heterogeneity analysis showed a significant and large *Q* statistic (*Q* (87) = 2,495.29, *p* < 0.001) based on a measure of weighted squared deviations. The *I*² index (*I*² = 96.57%, *p* < 0.001) indicated that almost 97% of the observed variance in effect size was explained by systematic differences across the studies. The effect sizes for each COM dimension showed substantial heterogeneity across the studies (capability *Q* = 836.04, *p* < 0.001; opportunity *Q* = 948.13, *p* < 0.001; motivation *Q* = 667.36, *p* < 0.001). The observed heterogeneity indicates the need to explore moderators (Lipsey & Wilson, 2001) such as COM dimension subtypes, study location, study design, and gender of participants.

4.2. Moderator Variables

We explored the influence of four moderator variables: COM dimension subtypes (capability: physical and psychological; opportunity: physical and social; and motivation: reflective and automatic), as well as study location, study design, and gender of participants.

4.2.1. COM Dimension Subtypes

The results of the moderator analysis comparing effect sizes between the COM dimension subtypes and water-use behavior is presented in Table 3. Average effect sizes varied significantly by the outcome variable within each of the COM dimension subtypes and between group *Q* tests. Studies assessing physical and psychological capability showed weak associations based on nonoverlapping 95% confidence intervals. However, only psychological capability had a significant association with water-use behavior. The individual predictor variables of physical and psychological capability revealed a varied association with water-use behavior (Table 4). For physical capability, income (*r* = 0.53, 95% CI [0.16, 0.89]) correlated significantly with water-use behavior (Table 4). Wealthier people have the money to buy water-saving devices, if only they wanted to, but

Table 3
Moderator Analysis of Effect Sizes for the COM Subtypes

COM dimension and subtype	<i>k</i>	<i>r</i>	95% CI		<i>z</i>	<i>p</i>
			Lower	Upper		
Capability						
Physical	10	0.08	−0.05	0.22	1.23	0.220
Psychological	21	0.14	−0.03	0.30	1.65	<0.001
Opportunity						
Physical	16	0.28	0.18	0.37	5.62	<0.001
Social	16	−0.07	−0.21	0.07	−0.99	0.320
Motivation						
Reflective	21	−0.15	−0.08	0.37	1.29	0.197
Automatic	4	0.13	−0.08	0.33	1.23	<0.001

Note. Capability *Q* (1) = 794.99, *p* < 0.001. Opportunity *Q* (1) = 930.61, *p* < 0.001. Motivation *Q* (1) = 620.80, *p* < 0.001, *k* = number of effect sizes. COM, capability, opportunity, and motivation.

Table 4
Moderator Analysis of Effect Sizes for the Predictor Variables of Each COM Subtype

COM dimension and subtype	Predictor type	k	r	95% CI		z	p
				Lower	Upper		
Capability Physical	Age	1	−0.35	−0.69	−0.01	−2.03	0.042
	Income	6	0.53	0.16	0.89	2.79	<0.001
	Involvement	1	0.18	−0.30	0.67	0.74	0.460
	Past actions	1	0.47	−0.02	0.95	1.89	0.058
	Use of water-efficient devices	1	0.59	0.09	1.09	2.31	0.021
Capability Psychological	Awareness	1	0.30	−0.18	0.79	1.23	0.218
	Education	5	0.47	0.09	0.84	2.45	<0.001
	Environmental concern	3	0.51	0.11	0.91	2.52	<0.001
	Feedback	1	0.86	0.36	1.36	3.39	<0.001
	Information	5	0.78	0.40	1.15	4.06	<0.001
	Knowledge	1	0.71	0.23	1.19	2.91	<0.001
	Perceived behavioral control	3	0.50	0.11	0.89	2.48	0.013
	Perception of water-efficient devices	2	0.47	0.05	0.89	2.18	0.029
Opportunity Physical	House ownership	1	0.51	0.01	1.01	1.99	0.045
	Household size	6	0.04	−0.34	0.43	0.21	0.835
	Household type	1	0.20	−0.29	0.70	0.79	0.429
	Humidity	1	0.02	−0.50	0.55	0.08	0.937
	Landscape choices	1	0.24	−0.26	0.74	0.94	0.349
	Location	2	−0.20	−0.64	0.23	−0.93	0.354
	Rebate	1	−0.22	−0.55	0.46	−0.86	0.389
	Restriction	1	0.10	−0.41	0.61	0.38	0.708
	Temperature	1	0.08	−0.45	0.60	0.28	0.779
Opportunity Social	Time	1	0.25	−0.57	0.43	−0.29	0.774
	Normative beliefs	2	0.23	−0.12	0.59	1.29	0.194
	Descriptive norms	1	0.17	−0.42	0.76	0.55	0.581
	Personal identity	1	−0.04	−0.55	0.46	−0.16	0.871
	Social identity	1	−0.17	−0.68	0.33	−0.68	0.500
	Social norms	10	−0.09	−0.38	0.36	−0.05	0.959
Motivation Automatic	Trust	1	0.22	−0.22	0.68	1.01	<0.001
	commitment	1	0.20	−0.25	0.64	0.87	0.383
Motivation Reflective	Habits	3	−0.21	−0.45	0.04	−1.65	0.098
	Attitudes toward water conservation	12	0.25	0.14	0.36	4.57	<0.001
	Intentions	3	0.21	−0.01	0.87	1.32	0.059
	Mindfulness	1	0.43	−0.01	0.87	1.88	0.059
	Moral obligation	1	−0.11	−0.49	0.28	−0.54	0.588
	Prompt	1	0.03	−0.39	0.45	0.13	0.897
	Purchase decisions	1	−0.13	−0.52	0.26	−0.65	0.517
	Values	2	−0.01	−0.30	0.28	−0.07	0.944

Note. Capability $Q(2) = 312.60$, $p < 0.001$. Opportunity $Q(2) = 207.48$, $p < 0.001$. Motivation $Q(2) = 451.66$, $p < 0.001$, k = number of effect sizes. COM, capability, opportunity, and motivation.

they do not want to; hence, they could do more to increase their intentions to purchase water-efficient devices. Lower income earners performed poorly in water-use behavior (i.e., water efficiency) presumably because they cannot afford to undertake these activities and they have limited intentions of buying water-efficient devices.

Water-use behavior (i.e., water efficiency) increased where people could use water-efficient devices, whereas water-use behavior was low when people failed to use these water-efficient devices (Table 4). Although use of water-efficient devices was significantly associated with water-use behavior, the strength of this association was moderate ($r = 0.59$, 95% CI [0.09, 1.09]). Capability (psychological) was measured in diverse ways;

Table 5
Moderator Analysis of Effect Sizes for Study Location

Study location	<i>k</i>	<i>r</i>	95% CI		<i>z</i>	<i>p</i>
			Lower	Upper		
Capability						
Australia	12	0.18	0.04	0.32	2.53	<0.001
Bulgaria	2	−0.02	−0.38	0.34	−0.12	0.905
China	3	−0.11	−0.42	0.19	−0.72	0.472
Spain	1	−0.17	−0.66	0.32	−0.67	0.502
Taiwan	1	0.02	−0.48	0.53	0.09	0.925
UK	2	0.07	−0.29	0.43	0.36	0.716
United States	10	0.04	−0.17	0.24	0.36	0.718
Opportunity						
Australia	8	0.33	0.21	0.46	5.13	<0.001
Botswana	1	−0.09	−0.45	0.28	−0.47	0.641
China	2	−0.25	−0.52	0.02	−1.79	0.073
Mexico	2	−0.30	−0.58	−0.02	−2.11	0.035
Spain	1	−0.24	−0.60	0.12	−1.30	0.192
Taiwan	1	−0.07	−0.45	0.32	−0.33	0.739
UK	3	0.19	−0.06	0.44	1.48	0.138
United States	14	−0.12	−0.28	0.03	−1.56	0.118
Motivation						
Australia	12	0.22	0.10	0.34	3.67	<0.001
Botswana	1	0.01	−0.42	0.44	0.05	0.959
Bulgaria	1	0.37	−0.06	0.79	1.69	0.090
China	1	−0.16	−0.59	0.26	−0.76	0.450
Taiwan	1	0.02	−0.42	0.48	0.10	0.924
UK	2	−0.10	−0.41	0.21	−0.62	0.573
United States	7	0.10	−0.11	0.30	0.93	0.352

Note. Capability $Q(3) = 594.32$, $p < 0.001$, Opportunity $Q(3) = 420.23$, $p < 0.001$, Motivation $Q(3) = 502.29$, $p < 0.001$, k = number of effect sizes.

of automatic motivation (and not the individual measures on their own) accounted for water-use behavior. Attitude toward water conservation was the only statistically significant measure of reflective motivation ($r = 0.25$, 95% CI [0.14, 0.36]), showing a weak positive association with water-use behavior (Table 4).

4.3. Study Location

For the majority of studies, study location did not significantly moderate effect sizes (Table 5). However, Australia was the exception. We compared studies conducted in Australia with those in Bulgaria,

however, all predictor variables except awareness ($r = 0.30$, 95% CI [−0.18, 0.79]) demonstrated perceived behavioral control ($r = 0.50$, 95% CI [0.11, 0.89]) and perception of water-efficient devices ($r = 0.47$, 95% CI [0.05, 0.89]) significantly moderated water-use behavior (Table 4). Psychological capability measures such as feedback ($r = 0.86$, 95% CI [0.36, 1.36]), information ($r = 0.78$, 95% CI [0.40, 1.15]), and knowledge ($r = 0.71$, 95% CI [0.23, 1.19]) strongly moderated water-use behavior, whereas education ($r = 0.47$, 95% CI [0.09, 0.84]) and environmental concern ($r = 0.51$, 95% CI [0.11, 0.91]) revealed a weak association with water-use behavior (Table 4).

The moderator analysis of the opportunity subtypes (i.e., physical and social) also showed a weak but significant association with water-use behavior. Physical opportunity was the only moderator that accounted for this significant association with water-use behavior (Table 3). Among the various physical opportunity measures, only house ownership revealed a moderate association with activities that promote water-use behavior. Although the average correlation of social opportunity ($r = −0.07$, 95% CI [−0.21, 0.07]) shows it is weak and not statistically significant, *trust* ($r = 0.22$, 95% CI [0.68, 1.01]) was the only significant social opportunity measure driving water-use behavior in a weak but positive linear direction (Table 4).

In terms of motivation, studies that focused on reflective and automatic motivation subtypes produced a weak but significant linear association with water-use behavior. Although 21 and 4 effect sizes were observed for reflective motivation and automatic motivation, respectively (Table 3), automatic motivation measures revealed a statistically significant association with water-use behavior ($r = 0.13$, 95% CI [−0.08, 0.33]). The association between automatic motivation and water-use behavior was, however, weak. It is worth noting that the interaction between measures

of automatic motivation (and not the individual measures on their own) accounted for water-use behavior. Attitude toward water conservation was the only statistically significant measure of reflective motivation ($r = 0.25$, 95% CI [0.14, 0.36]), showing a weak positive association with water-use behavior (Table 4).

Table 6
Moderator Analysis of Effect Sizes for Study Design

Study design	<i>k</i>	<i>r</i>	95% CI		<i>z</i>	<i>p</i>
			Lower	Upper		
Capability						
Correlational	16	0.19	0.13	0.24	6.35	<0.001
Experimental	4	0.16	0.13	0.58	2.47	<0.001
Longitudinal	11	0.05	−0.05	0.16	1.06	0.290
Opportunity						
Correlational	20	0.23	0.15	0.32	5.18	<0.001
Experimental	5	0.02	−0.20	0.23	0.15	0.879
Longitudinal	7	0.03	−0.15	0.20	0.32	0.749
Motivation						
Correlational	14	0.22	0.11	0.33	3.98	<0.001
Experimental	6	0.12	−0.09	0.34	1.12	0.262
Longitudinal	5	0.01	−0.20	0.23	0.13	0.898

Note. Capability $Q(4) = 705.04$, $p < 0.001$. Opportunity $Q(4) = 940.35$, $p < 0.001$. Motivation $Q(4) = 641.92$, $p < 0.001$, k = number of effect sizes.

Botswana, China, Mexico, Spain, Taiwan, UK, and United States. Group Q tests revealed significant differences between studies in these countries for all the COM dimensions. Studies conducted in Australia (e.g., $r_{\text{capability}} = 0.18$, 95% CI [0.04, 0.32]) revealed a significant but weak linear association with water-use behavior (Table 5). However, participants in studies conducted in Australia and Mexico used both the physical and social opportunity dimensions to achieve water-use behavior. Studies from Australia ($r_{\text{opportunity}} = 0.33$, 95% CI [0.21, 0.46]) showed that opportunity measures and water-use behavior were weakly associated in the same direction, whereas in Mexico the association was inversely correlated ($r_{\text{opportunity}} = −0.30$, 95% CI [−0.58, −0.02]).

4.3.1. Study Design

Moderator analysis showed variance in results across study design subtypes (i.e., correlational, experimental, and longitudinal) for all the COM dimensions (Table 6). Experimental and correlational studies weakly influence effect sizes in a positive direction, whereas no-significant influence was observed for longitudinal studies (Table 6). Correlational studies had a significant but a weak influence on effect sizes (Table 6). Studies

Table 7
Moderator Analysis of Effect Sizes for the Gender of Participants

Gender	<i>k</i>	<i>r</i>	95% CI		<i>z</i>	<i>p</i>
			Lower	Upper		
Capability						
Male	8	0.23	−0.20	0.36	3.28	<0.001
Female	8	0.41	0.04	0.48	3.06	<0.001
Opportunity						
Male	14	−0.25	−0.36	0.32	4.64	<0.001
Female	14	0.25	0.14	0.35	4.64	<0.001
Motivation						
Male	13	0.26	0.14	0.39	4.11	<0.001
Female	14	0.30	0.19	0.42	5.20	<0.001

Note. Capability $Q(5) = 835.41$, $p < 0.001$. Opportunity $Q(5) = 902.98$, $p < 0.001$. Motivation $Q(5) = 632.91$, $p < 0.001$, k = number of effect sizes.

involving correlational design were the only measure that was significantly associated with study effect sizes in a positive linear direction (e.g., motivation $r_{\text{correlational}} = 0.22$, 95% CI [0.11, 0.33]), indicating that studies conducted using a correlational approach were related to an increase or decrease in household water use.

4.3.2. Gender of Participants

Most studies included in the meta-analysis did not report gender, thereby reducing effect sizes to $k = 36$. However, moderator analysis showed significant heterogeneity ($p < 0.001$) across gender for all the COM dimensions and the water-use behavior subtypes (Table 7). Studies with more female participants had stronger water-use behavior measure effect sizes than those with more male participants. The capability of people to promote water-conservation behavior was gender insensitive as weak linear association was observed for both sexes: male ($r_{\text{male}} = 0.23$, 95% CI [−0.20, 0.36]) and female ($r_{\text{female}} = 0.41$, 95% CI [0.04, 0.48]). For opportunity, male

participants ($r_{\text{male}} = -0.25$, 95% CI [−0.36, 0.32]) showed a significant but weak negative influence on effect sizes whereas female participants ($r_{\text{female}} = 0.25$, 95% CI [0.14, 0.35]) revealed a positive influence (Table 7). There was no difference when studies were split by gender as both male and female participants showed that effect sizes for motivation measures were significantly associated with water-use behavior in a positive linear direction for males ($r_{\text{male}} = 0.26$, 95% CI [0.14, 0.39]) and females ($r_{\text{female}} = 0.30$, 95% CI [0.19, 0.42]).

5. Discussion

The meta-analysis examined associations between scores on COM dimensions and household water-use behavior by incorporating the outcomes of studies that have both psychological-social predictors of water-use behavior. Higher scores on COM dimensions were associated with higher levels of perceived household water-use behavior. Moderation analyses showed correlation coefficients to vary significantly as a function of COM dimension subtypes, study location, study design, and gender. Positive correlations between COM dimensions and water-use behavior shows when an increase in an opportunity dimension measure occurs (e.g., water-use restrictions), a corresponding increase in the tendency for water-use reduction should be expected. Negative correlations also occurred when variables move in an inverse or opposite direction. Negative correlations between opportunity dimension measures (e.g., rebates) and water-use behavior mean that when a rebate increases, the possibility of reducing water consumption is achievable because households will have greater incentives to buy water-efficient appliances. This correlational strength or direction applies to the individual measures (predictor variables) of the COM dimensions.

5.1. COM Dimensions for Understanding Household Water-Use Behavior

Moderate effect sizes for all the COM dimensions suggest water-saving activities are consistent with the BCW model, and COM dimensions influence water-conservation behavior. The results showed that psychological-social predictors (e.g., environmental concern, education, income, feedback, house ownership, perceived behavioral control, trust, information, and positive attitudes toward conservation) had modest effects on defined water outcome measures (e.g., water-saving activities), which may include garden watering, showering times, metered household consumption, self-reported water use, curtailment, and efficiency behaviors. All three dimensions of the BCW framework were hypothesized to be significantly correlated with household water-use behavior in the meta-analysis. The BCW framework suggested that respondents who scored high on the COM dimensions were inclined to change behavior. According to the BCW framework, behavior is part of an interacting system of behavioral conditions required to provide targets for intervention (Michie et al., 2011). The effectiveness or strength of the correlational association between the COM dimensions and household water-use behavior varied significantly as a function of the moderator variables of each COM dimension.

5.2. Moderator Analyses for each COM Dimension on Water-Use Behavior

The effectiveness or direction of the correlational association between the COM dimensions and household water-use outcome measures varied significantly as a function of the moderators (e.g., feedback, information, environmental concern, and knowledge) and COM dimension subtypes, study location, and gender participants.

The moderator analyses revealed that the capability dimension subtype of psychological capability had the strongest effect on household water-use behavior. This was associated with feedback, information, environmental concern, and knowledge, highlighting that a households' logical reasoning and understanding to engage in the necessary mental processes occurred sufficiently, or partially, with respect to water-conservation activities. For example, understanding the impact of greenhouse emissions on the environment was more important than having the physical skills, strength, or stamina to engage in water-use reduction activities (e.g., check and repair minor water leakages). The significant effect size for psychological capability suggests that personal experience with water shortages can facilitate conservation activities. For example, having experienced a water crisis provides knowledge to adapt to water shortages. In the semiarid areas of China, for example, educated farmers (i.e., having psychological capability) were more likely to use rain-water harvesting and supplementary irrigation technology to alleviate water shortages (He et al., 2007).

Participants who scored higher on the capability dimension perceived vital information for strategic conservation activities by highlighting which activities require critical psychological skills to engage in conservation measures. This insight aids a household to strive to improve their skills in order to reduce the risk of water shortages. To do so they enhance their selected skills based on the capabilities needed to be successful in conservation behaviors. Consistent with the findings of Russell and Fielding (2010), households with psychological knowledge may exhibit high awareness of water conservation intentions and install water-efficient appliances to reduce water consumption. Our findings support the proposition that water-conservation programs should relate to behavior conditions to strengthen and promote extensive water-conservation practices within localities. For example, government water agencies working in areas prone to water scarcity may understand that water-use behaviors can differ in context, drivers, and barriers and provide proactive interventions that improve coping responses and increase resilience within their communities. This could involve providing the community with capability-focused strategies that focus on individuals' physical and psychological abilities to enact the desired behavior (e.g., feedback, information related to water-efficient devices, and skills support).

The social opportunity dimension displayed an inverse correlation in our meta-analysis, albeit nonsignificant (Table 3). This was based on social opportunity measures in driving water-use behavior. However, the meta-analysis did show trust to have a significant effect. The significant correlation between trust and water-use behavior for households is consistent with the assertions of the BCW that households are committed to conserving water when interpersonal influences (social opportunity) such as trust, social cues, and cultural norms affect the way they think about water resources. Richter (2014) argues that water-conservation practices should be encouraged through cultural beliefs and interpersonal trust. This provides an avenue for water authorities and governments to establish trust in delivering quality services to water users (Jorgensen et al., 2009). In addition, awareness campaigns and water-conservation programs should be tailored to the beliefs of the community. Tailoring awareness campaigns and water-conservation programs to beliefs of the community have turned out to be striking, constructive, and cost-effective in various spheres of water crises (Gilli, 2004).

Motivation subtypes observed higher intentions and emotional reactions with water-use behavior than those who obtained lower scores on motivation subtypes. This suggests motivation is moderately related to water-use behavior. The significant correlation between automatic motivation measures (e.g., habits) and water-use behavior may imply an individual's ethical drive to reduce consumption. Automatic habits or routines (Steg & Vlek, 2009) are practiced behavioral tendencies generating a belief among householders that water crises create a possible hazard to society by increasing socio-economic uncertainty, poor productivity, health issues, and social injustice (Memon & Butler, 2006; Richter, 2014). This association is partly due to the relationship between an individual's high voluntary water-conservation habits and perceived future water crises that may make households susceptible to more bouts of water shortages. In contrast, other repercussions of water crises, such as warfare, sanitation issues, hunger, reduced economic activities, and poverty, are more noticeable and ascend the need to discourage the occurrence of water shortages. Therefore, motivated householders may habitually respond to water-conservation practices because they can voluntarily conserve water without complying with any restrictions and rely on available information related to water crises and conservation behavior.

Moderator analyses for study location showed that most studies assessing associations between COM dimensions and water-use behavior have occurred in Australia, UK, and the United States. Australian studies, on

average, produce larger effect sizes. This result suggests that the more susceptible a country is to a water crisis, the larger the effect size, which in turn encourages and strengthens efforts to sustain water resources. The majority of water-conservation behavior studies included in the meta-analysis occurred in Australia, reflecting the prevalent water issues in this arid country (Australian Bureau of Statistics, 2014). The Australian Government and water industries are gradually improving the implementation and enforcement of water-demand management strategies and environmental policy such as water metering, water restrictions, pricing structure, and retrofitting in order to make household water-use behavior possible or to prompt it. Overall, these factors proved to be less reliable and presented smaller and weaker assenting strength in non-Australian cases. Thus, it is not unexpected that the COM dimensions employed in this study generated significantly greater correlation coefficients in Australia.

Moderate effect sizes were found in capability, opportunity and motivation, and the gender of participants. Thus, gender is highly associated with water-conservation behavior. For opportunity, both men and women perceived water-use behavior, but in opposite directions. Women habitually conserve water and have a strong concern for the efficiency and sustainability of water interventions and the management of water resources. Studies on gender differences in pro-environmental behavior validate that women intuitively report engaging in more proenvironmental behaviors than their male counterparts (Lam & Cheng, 2002; Zelezny et al., 2000). Women show greater positive attitudes toward water conservation, valuing conservation practices, expressing interest in collaborating with water authorities, and relying on conservation information more than men (Druschke & Secchi, 2014). Thus, activities related to household water reduction should be more orientated to women.

5.3. COM Dimensions: Implications for Water Conservation

The accuracy of using the COM dimensions as sources of behavior change is important if they are used as the principal variables in intervention development. It should be perceived as a novel behavior-change model providing interventions whose effects are more answerable, transparent, mediated, and moderated by other behavioral factors. This suggests that linking COM dimensions and targeted household water-use behaviors can provide specific conservation intervention strategies of which individuals will be able to recognize to adopt and promote worthwhile household water-conservation practices. The BCW framework assumes that behavioral change is determined by three factors: capability (determined by physical and psychological skills), opportunity (characterized by physical and social factors), and motivation (perceived by reflective and automatic reactions), which can be used for understanding behavior and predicting behavior change. The COM dimensions provide excellent and precise predictions of household water behavioral feedback within an applied frame of reference. It shows whether individual households will be capable of being actively involved in water conservation in the face of climate variations such as drought. Water conservation practitioners and householders can apply these findings to modify outreach applications that will further enhance community water-conservation programs.

5.4. Research Limitations

There were various constraints associated with the interpretation of the results of this meta-analysis. Distinct categories covered in the moderator analyses comprised comparatively small effect sizes. In particular, average effect sizes from studies conducted in Botswana, Bulgaria, Taiwan, UK, and the United States should be treated with caution. One possible explanation for this was that insufficient information was available for the moderation analyses and the sample sizes were not large enough because not every study generated the data necessary for moderation analyses. Additionally, the rigid inclusion criteria to establish precision of the analyses and significance of the outcomes suggested that some valuable information (e.g., data in conference proceedings, seminal, and unpublished studies) was inappropriate for the analysis. Similarly, the current investigation focused on household water use as opposed to other water uses, for example, in the agricultural and industrial sectors. Although many of the studies deliberated about these water uses, measures that were provided in the literature were limited to household water use. Given the inverse associations between diverse predictor variables and conservation approaches, it gives the impression that multiple conservation behaviors may not concurrently influence households' water-use behavior, the expression of which may be location-specific. If this is accurate, it may be practical to conduct empirical studies based on different predictors and water-use behaviors and investigate the divergent practices in which water-use behaviors are related.

Our research raises questions for further investigation. First, future research might strive to classify the locations under which COM dimensions have the strongest and least effectiveness, which would suggest when these sources of behavior may be most reliable and applicable. Our investigation revealed the need for rigorous studies in multiple settings to determine the extent to which COM dimensions could be applied to well-defined pro-environmental behaviors in all sectors. Studies have found evidence that daily environmental actions change gradually over time, but the effect of pro-environmental behaviors in conserving resources demands activities be considered to accelerate conservation measures (Dascher et al., 2014; Goldsmith & Goldsmith, 2011; Wahlen, 2011). Second, using mixed method approaches may help identify the most appropriate sources of behavior for promoting specific conservation strategies among all water users. Anticipating this research, policy-makers, practitioners, and water authorities should note that there are many influences on water-use behavior and a homogenous approach to promoting water conservation may be less successful.

6. Conclusions

The results of our meta-analysis on water-use behaviors suggest that household water-use behavior is associated with three conditions: capability, opportunity, and motivation. As hypothesized by the BCW framework, we found that households may adopt water conserving behaviors when they have a higher perception of being capable, when having all the necessary opportunities at their disposal, and when motivated to conserve water. Moderator analyses showed that correlation coefficients differed significantly with COM dimension subtypes, study location, study design, and gender of participants. Overall, the results of our study are generally aligned with the BCW framework's assertion that the integrative components of behavior are important sources of the psychological-social predictors of household water-use behavior. This highlights the importance of viewing the COM dimensions as hypothetical sources of psychological-social predictors that water utilities, practitioners, governments, and stakeholders might use to promote water-conservation behaviors. However, while the COM dimensions are useful, specific results may diverge considerably based on the water-use characteristics of a region.

It was apparent from the meta-analysis that household water conservation behavioral change and understanding the mechanisms of behavioral change encourages long-term household water-conservation. A focus on the direct causal mechanisms of long-term water conservation behavior in households is an alternative approach to traditional command-and-control approaches to water conservation. Command-and-control strategies are activities directly regulated by a governing entity that states what is permitted and what is illegal (McManus, 2009) with negative consequences for noncompliance such as fines or prosecution. Command-and-control strategies have been shown to have limited long-term benefit to conserving water because they do not encourage sustained behavior change (McGranahan, 2002). Understanding and changing behavior is one of the most important and emerging alternatives for conserving water in the 21st century (Gleick, 2003; Hoque, 2014), and this study shows how water-conservation outcomes can be improved through knowledge of the psychological-social drivers of water conservation behavior.

This is the first meta-analysis to our knowledge that focuses on causal mechanisms of water-use behavior. Organizing psychological-social drivers and their effects on water-use behavior into a broad integrative model of behavior comprising capability, opportunity, and motivation dimensions allows for the identification of corresponding interventions such as fiscal measures, guidelines, environmental/social planning, communication/marketing, legislation, service provision, and regulation. Effective water conservation strategies must relate water-use behavior to its causal mechanisms and interventions. If this is not done the link between variable and effect fails, because interventions target the wrong behaviors (Bartholomew et al., 2011). Bandura and Watts (1996) noted that a lack of linking appropriate household water-use behavior to relevant interventions affected the success of these interventions. Addressing the issue of the causal mechanisms of psychological-social drivers of water-conservation using the mechanisms of COM is a prerequisite to understanding and changing attitudes and behaviors to water conservation.

A major implication of the meta-analysis findings is that women are more capable of conserving water. Further, households with greater income and higher levels of education are likely to adopt water saving measures. Thus, knowing something about these demographic characteristics of communities will enable more efficient targeting and therefore better outcomes of water conservation strategies, particularly when combined with understanding of the COM dimensions of behavior change.

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